KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)
Deemed to be University U/S 3 of UGC Act, 1956

CAPSTONE PROJECT REPORT

Implementation of Procure-to-Pay (P2P) Cycle in Supply Chain Management (SAP MM)

Submitted By: URVI SAYTA
Roll No: 2306075
Program: B.Tech (INFORMATION TECHNOLOGY)
Course: SAP BTP Developer
Semester: 6th
College: KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT)

INDEX

1. Introduction
2. Company Scenario
3. Organizational Structure
4. Master Data
5. Procure-to-Pay Process
6. Flowchart
7. Accounting Entries
8. Advantages
9. Conclusion

1. Introduction

The Procure-to-Pay (P2P) cycle is a key process in supply chain management that integrates purchasing, inventory, and finance functions within an organization. It involves the complete lifecycle of procuring goods from external vendors and making payments using SAP MM.

This process ensures efficient procurement planning, proper inventory control, and accurate financial transactions. It also helps organizations maintain transparency, reduce errors, and improve operational efficiency.

2. Company Scenario

ABC Pvt Ltd is a manufacturing company that procures raw materials from external vendors to produce finished goods. The company uses SAP MM to manage its procurement activities efficiently.

Whenever there is a requirement for raw materials, the procurement department initiates the purchasing process, ensuring timely availability of materials while maintaining cost efficiency and proper vendor management.

3. Organizational Structure

The organizational structure defines how different business units are arranged within SAP:

- **Company Code:** ABCD (Represents the legal entity)
- **Plant:** PL01 (Location where goods are manufactured/stored)
- **Storage Location:** SL01 (Place where inventory is physically stored)
- **Purchasing Organization:** PO01 (Responsible for procurement activities)

This structure helps in proper coordination of procurement, inventory, and financial processes.

4. Master Data

Master data is essential for executing procurement processes in SAP:

- **Material Master (MM01):**

  Contains details about materials such as description, unit, and valuation.

- **Vendor Master (XK01):**

  Stores information about suppliers including name, address, and payment terms.

- **Purchasing Info Record (ME11):**

Maintains the relationship between vendor and material, including pricing and delivery conditions.

Proper master data ensures smooth and accurate execution of the P2P process.

5. Procure-to-Pay Process

Step 1: Purchase Requisition (ME51N)

A Purchase Requisition is an internal document created to request the procurement of materials or services. It is created using transaction ME51N. The requisition includes details like material, quantity, required date, plant (PL01), storage location (SL01), and purchasing organization (PO01).

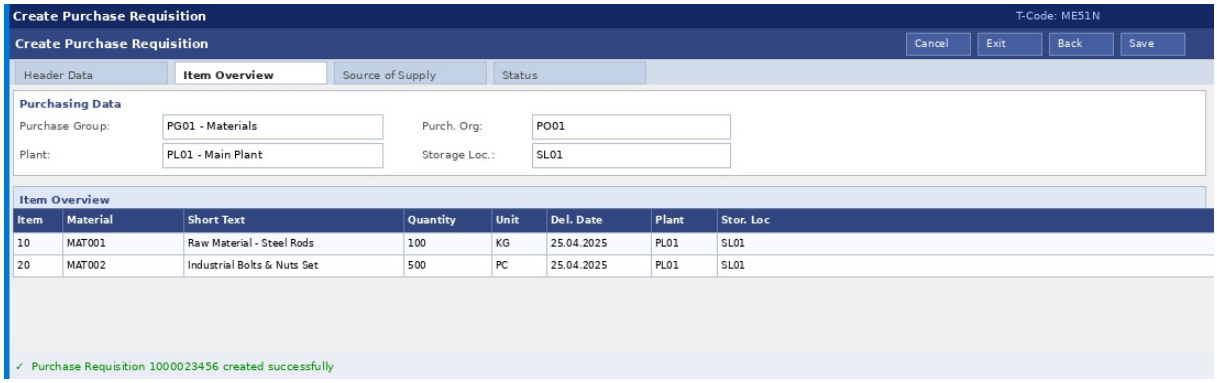


Figure 1: Purchase Requisition Creation (ME51N)

Step 2: Purchase Order (ME21N)

A Purchase Order (PO) is a formal document sent to the vendor to procure goods or services. It is created using transaction ME21N with reference to the purchase requisition. The PO contains vendor details (XYZ Traders), material info (LAPTOP - PR01), quantity (50), and value (5,000.00).

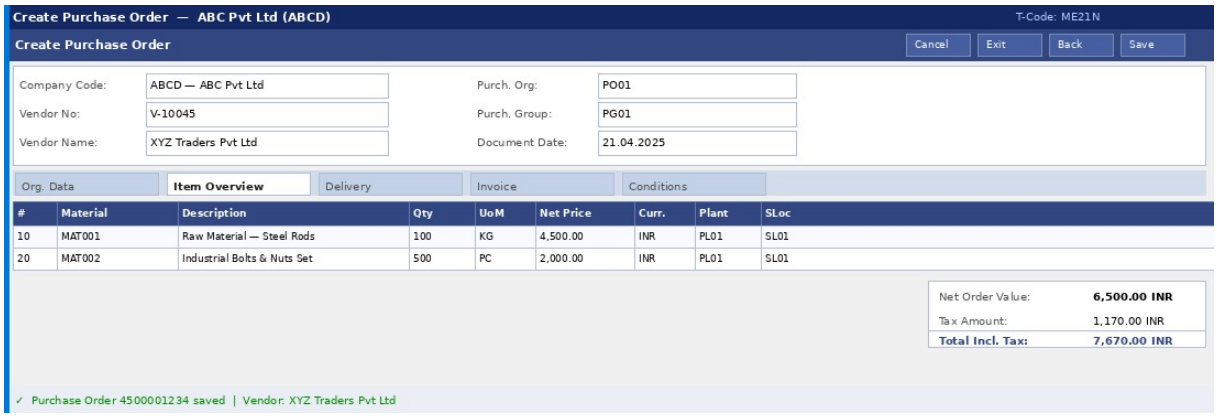
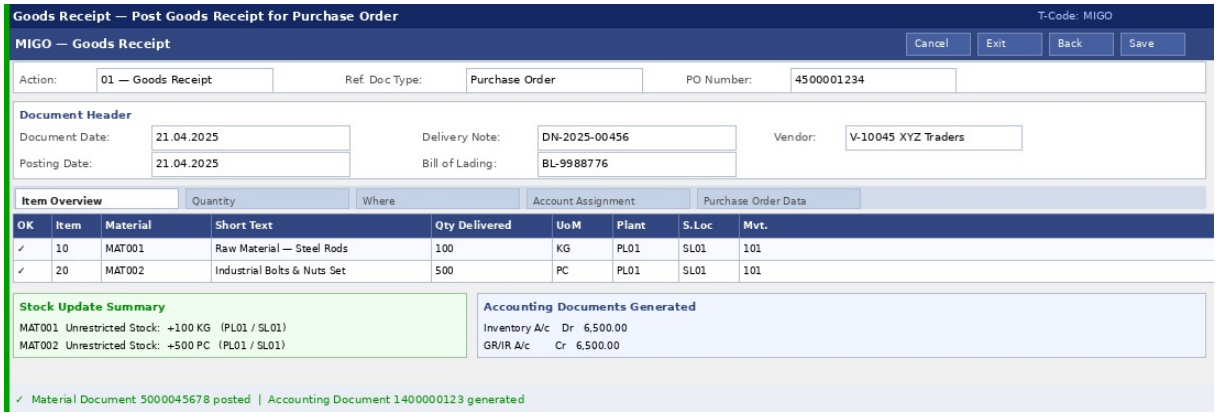


Figure 2: Purchase Order Creation (ME21N)

Step 3: Goods Receipt (MIGO)

When goods are physically received from the vendor, a Goods Receipt (GR) is posted using transaction MIGO. This updates inventory and triggers the financial accounting entry. The GR references the PO and records the actual quantity received at the plant.



Goods Receipt — Post Goods Receipt for Purchase Order T-Code: MIGO

MIGO — Goods Receipt Cancel | Exit | Back | Save

Action: 01 — Goods Receipt Ref. Doc Type: Purchase Order PO Number: 4500001234

Document Header

Document Date: 21.04.2025 Delivery Note: DN-2025-00456 Vendor: V-10045 XYZ Traders

Posting Date: 21.04.2025 Bill of Lading: BL-9988776

Item Overview | Quantity | Where | Account Assignment | Purchase Order Data

OK	Item	Material	Short Text	Qty Delivered	UoM	Plant	S.Loc	Mvt.
✓	10	MAT001	Raw Material — Steel Rods	100	KG	PL01	SL01	101
✓	20	MAT002	Industrial Bolts & Nuts Set	500	PC	PL01	SL01	101

Stock Update Summary

MAT001 Unrestricted Stock: +100 KG (PL01 / SL01)

MAT002 Unrestricted Stock: +500 PC (PL01 / SL01)

Accounting Documents Generated

Inventory A/c Dr 6,500.00

GR/IR A/c Cr 6,500.00

✓ Material Document 5000045678 posted | Accounting Document 1400000123 generated

Figure 3: Goods Receipt (MIGO)

Step 4: Invoice Verification (MIRO)

Invoice Verification is done using transaction MIRO. The vendor's invoice is matched against the Purchase Order and Goods Receipt (3-way match). This step ensures that payment is only processed for goods actually received and at the agreed price.

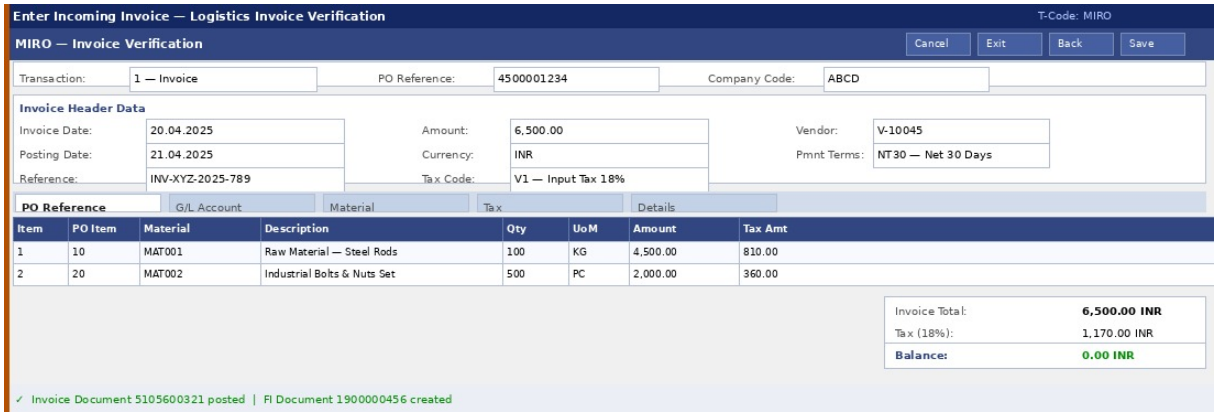


Figure 4: Invoice Verification (MIRO)

Step 5: Payment (F-53/F110)

The final step is making payment to the vendor. This can be done manually using transaction F-53 (Post Outgoing Payments) or automatically via F110 (Automatic Payment Program). Payment details include bank account, amount, document date, and vendor reference.

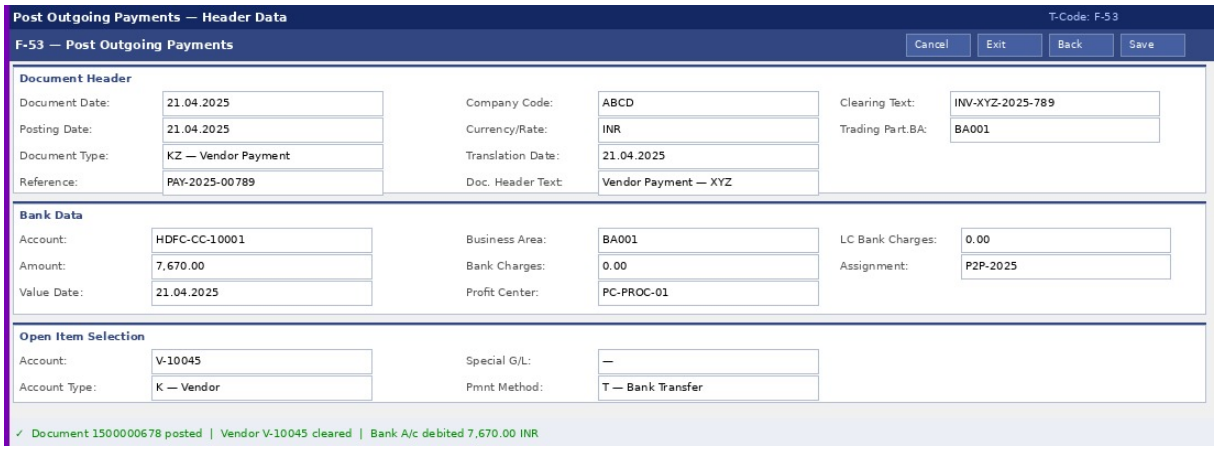


Figure 5: Payment Processing (F-53/F110)

6. Flowchart

PR → PO → GR → Invoice → Payment

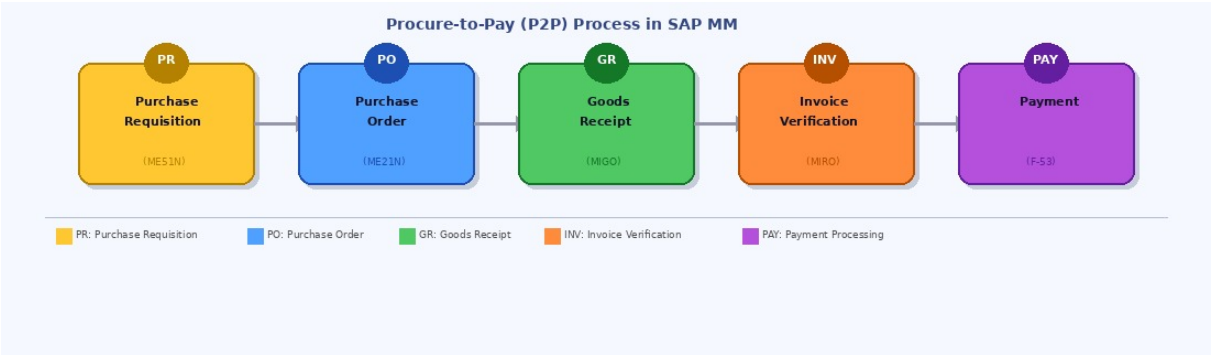


Figure 1: Procure-to-Pay (P2P) Process in SAP PR: Purchase Requisition • PO: Purchase Order • GR: Goods Receipt • INV: Invoice Verification

7. Accounting Entries

The Procure-to-Pay process is integrated with Financial Accounting (FI), ensuring that all transactions are recorded accurately.

1. Goods Receipt (MIGO)

When goods are received, inventory is increased:

```
Inventory A/c Dr To GR/IR A/c
```

2. Invoice Verification (MIRO)

When the vendor invoice is posted:

```
GR/IR A/c Dr To Vendor A/c
```

3. Payment to Vendor (F-53 / F110)

When payment is made:

```
Vendor A/c Dr To Bank A/c
```

8. Advantages

The Procure-to-Pay process provides several benefits to organizations:

- **Automation:** Reduces manual work and speeds up procurement
- **Accuracy:** Minimizes errors through system validation
- **Transparency:** Provides clear visibility of all transactions
- **Better Vendor Management:** Tracks vendor performance and payments
- **Cost Control:** Helps in monitoring and reducing procurement costs
- **Integration:** Connects procurement with inventory and finance modules

9. Conclusion

The Procure-to-Pay (P2P) cycle in SAP MM plays a crucial role in managing procurement activities efficiently. It integrates purchasing, inventory management, and financial accounting into a single system.

By automating the procurement process, SAP ensures accuracy, transparency, and better control over business operations. The P2P cycle helps organizations improve efficiency, maintain proper records, and build strong relationships with vendors.